METRIC

MIL-DTL-24521B(SH)
18 October 2022

SUPERSEDING
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DETAIL SPECIFICATION

SODIUM NITRITE, TECHNICAL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers technical-grade sodium nitrite for use as an anti-corrosive agent in boiler preparation.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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AMERICAN CHEMICAL SOCIETY (ACS)

ACS Reagent Chemicals - Specifications and Procedures for Reagents and Standard-Grade Reference Materials

(Copies of this document are available online at https://pubs.acs.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQ H1331 - Zero Acceptance Number Sampling Plans

(Copies of this document are available online at https://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM D1193 - Standard Specification for Reagent Water

ASTM D1293 - Standard Test Methods for pH of Water

ASTM E287 - Standard Specification for Laboratory Glass Graduated Burets

ASTM E288 - Standard Specification for Laboratory Glass Volumetric Flasks

ASTM E969 - Standard Specification for Glass Volumetric (Transfer) Pipets

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Sodium nitrite shall be in the form of anhydrous crystals and conform to the formula $NaNO_2$.

3.2 Chemical requirements. The sodium nitrite shall conform to the chemical requirements specified in table I when tested as specified in 4.2.

TABLE I. Chemical requirements.

Constituent or property	Value	Percent by weight	Test paragraph
Sodium nitrite	----	97.0 (minimum)	4.2.3
Sulfates (as Na_2SO_4)	----	0.2 (maximum)	4.2.4
Chlorides (as NaCl)	----	0.2 (maximum)	4.2.5
Insoluble matter	----	0.5 (maximum)	4.2.2
pH value	8±1	----	4.2.6

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions specified herein. The characteristics specified in table I, when tested in accordance with 4.2, shall constitute the minimum inspections to be performed. For purposes of sampling, an inspection lot shall consist of all material submitted for inspection and delivery at one time. Unless otherwise specified (see 6.2), sampling shall be in accordance with MIL-STD-1916, ASQ H1331, or a comparable zero-based sampling plan. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

4.2 Test methods. Unless otherwise specified (see 6.2), the physical and chemical values specified in 3.1 and table I shall apply to the average of two determinations made on the sample of sodium nitrite.

4.2.1 Test preparation. Reagent water conforming to type III grade of ASTM D1193 and reagent chemicals conforming to ACS Reagent Chemicals shall be used throughout the tests. Laboratory glassware used in testing shall be in accordance with ASTM E287, ASTM E288, and ASTM E969. Weighings shall be made to the nearest 0.0001 gram. Sodium nitrite shall be weighed as received. Unless otherwise specified herein, solutions shall be made up on a weight basis.

4.2.2 Insoluble matter test. For insoluble matter, the following procedure shall be used:

a. Dissolve 25 grams of sample in 250 milliliters of water, heat to boiling, and digest in a covered beaker on a steam bath for 1 hour.

b. Filter the liquid through a tared sintered-glass crucible (fine porosity with a nominal maximum pore diameter of 5 micrometers) that has been dried to constant weight at 110±2 °C.

c. Quantitatively transfer insoluble matter to the crucible by washing the beaker with hot reagent water.

d. Retain the combined filtrate and wash water for the sulfate test (see 4.2.4) and chloride test (see 4.2.5).

e. Dry the filtering crucible to constant weight at 110±2 °C.

f. Calculate the percent insoluble matter as follows:

$$Percent\ insoluble\ matter = \frac{100(A - B)}{W}$$

Where:

A = Weight of crucible plus residue in grams

B = Weight of crucible in grams

W = Weight of sample in grams

4.2.3 Sodium nitrite test. For sodium nitrite, the following procedure shall be used:

a. Dissolve 10 grams of sample in water and dilute to 1,000 milliliters in a volumetric flask.

b. Add 5 milliliters of concentrated sulfuric acid to 300 milliliters of water in a 600-milliliter beaker.

c. Stir, and while the solution is still warm, add 0.1N potassium permanganate until a faint pink color is produced that persists for 2 minutes.

d. Add 40 milliliters of 0.1N potassium permanganate and stir gently.

e. Pipet 10 milliliters of the diluted sample, slowly and with constant agitation, into the solution in the 600-milliliter beaker.

f. Keep the tip of the pipet well below the surface of the liquid.

g. Add 15 milliliters of 0.1N ferrous ammonium sulfate and allow the solution to stand for 5 minutes.

h. Titrate the excess of ferrous ammonium sulfate with 0.1N potassium permanganate and record this value as "A".

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i. Calculate the percent sodium nitrite as follows:

$$Percent\ sodium\ nitrite = \frac{0.345A}{W}$$

Where:

A = Milliliters of 0.1N potassium permanganate used to back titrate

W = Weight of sample in aliquot in grams

4.2.4 Sulfate test. For sulfate, the following procedure shall be used:

a. Cool the filtrate and washings retained from 4.2.2.

b. Quantitatively transfer the solution to a 500-milliliter volumetric flask and dilute to volume.

c. Pipet 200 milliliters of this solution into a 400-milliliter beaker.

d. Add methyl orange indicator.

e. Add sufficient hydrochloric acid to make the solution just acidic to methyl orange.

f. Add 2 milliliters of concentrated hydrochloric acid in excess.

g. Heat to boiling.

h. Add dropwise, while stirring, 10 milliliters of 10 percent barium chloride solution.

i. Continue boiling for 5 minutes.

j. Remove the solution from the hot plate and allow the precipitate to settle.

k. Filter the solution through a tared sintered-glass crucible (with a nominal maximum pore diameter of 5 micrometers) that has been dried to constant weight at 110±2 °C.

l. Quantitatively transfer the precipitate to the filter.

m. Wash with hot reagent water until the filtrate is free of chloride as indicated by the absence of a precipitate in a silver nitrate solution.

n. Dry the filtering crucible to constant weight at 110±2 °C.

o. Calculate the percent sodium sulfate as follows:

$$Percent\ sodium\ sulfate = \frac{60.9(A - B)}{W}$$

Where:

A = Weight of precipitate plus crucible in grams

B = Weight of crucible in grams

W = Weight of sample in aliquot in grams

4.2.5 Chloride test. For chloride, the following procedure shall be used:

a. Pipet 200 milliliters of sample from the solution prepared from the filtrate and washings of 4.2.2 into a 400-milliliter beaker.

b. Add 5 milliliters of concentrated nitric acid.

c. Boil gently for 5 minutes.

d. Add 10 milliliters of approximately 0.1N silver nitrate dropwise with constant stirring and continue boiling for an additional 5 minutes.

e. Remove solution from the hot plate.

f. Allow the precipitate to settle.

g. Test the supernatant liquid with a few drops of 0.1N silver nitrate for completeness of precipitation.

h. If precipitation is not complete, add an additional 5 milliliters of 0.1N silver nitrate, boil as indicated above, and retest the supernatant.

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i. When precipitation is complete, filter the solution through a tared sintered-glass crucible (with a nominal maximum pore diameter of 5 micrometers) that has been dried to constant weight at 110±2 °C.

j. Quantitatively transfer the precipitate to the filter.

k. Wash with hot reagent water.

l. Use 1:100 nitric acid for transferring and washing of the precipitate.

m. Dry the filtering crucible to constant weight at 110±2 °C.

n. Calculate the percent sodium chloride as follows:

$$Percent\ sodium\ chloride = \frac{40.8(A - B)}{W}$$

Where:

A = Weight of crucible plus precipitate in grams

B = Weight of crucible in grams

W = Weight of sample in aliquot in grams

4.2.6 pH value test. For pH value, the following procedure shall be used:

a. Dissolve 2±0.005 grams of sample into 200 milliliters of neutral reagent water.

b. Determine the pH value of the solution in accordance with ASTM D1293.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sodium nitrite covered by this specification is intended for use as a corrosion inhibitor when preparing a boiler for either wet or dry lay-up in accordance with NAVSEA Standard Item 009-116. It is also used as a passivator following boiler chemical cleaning in accordance with MIL-STD-796 and MIL-STD-1607. It is used in water lancing of boilers in accordance with NAVSEA Standard Item 009-31. Current stock of sodium nitrite may be used until depleted; future requisitions for sodium nitrite should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required in pounds.

c. When sampling requirements are other than specified (see 4.1).

d. When number of determinations for average of values differs (see 4.2).

e. Packaging requirements (see 5.1).

f. Requirements for an SDS (see 6.3).

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6.3 Safety data sheets (SDS). As specified (see 6.2), the contracting activity will be provided an SDS at the time of contract award. The SDS will be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200. When FED-STD-313 is at variance with the Code of Federal Regulations (CFR), 29 CFR 1910.1200 will take precedence, modify, and supplement FED-STD-313. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313 and 29 CFR 1910.1200.

6.4 Subject term (key word) listing.

Chloride

Corrosion inhibitor

Insoluble matter

Oxidant

pH

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
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